

FINAL EXAM—SOLUTIONS

ECO403/503—INTRODUCTION TO CONTRACT THEORY AND AUCTIONS

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Instructions. The exam has a total of 120 points. *I will grade it over 100, so you can obtain an extra 20 points if you do everything right.* All questions have the same weight (40 points). Read the problems carefully; then decide which ones you want to attempt first. I suggest that you try to answer first the problems which you feel most confident about and then move on to the ones in which you don't feel so confident. Write your answers neatly and cross anything that you don't want me to consider. Please number each page you use. Scan your papers and upload them to Moodle in pdf format. The exam has a duration of 3 hours plus 30 minutes to scan and send your material.

1. Assume that there is a continuum of types of consumers, where their type is given by θ , which is Uniformly distributed on the interval $[\theta_L, \theta_H]$. A type's θ utility when consuming quantity x and paying an amount of T is given by

$$u(x, T, \theta) = \frac{\theta(1 - (1 - x)^2)}{2} - T$$

A firm is the sole producer of good x , and its cost of production per unit is c . In this question, you will find the optimal quantity-transfer scheme for each type, $(x(\theta), T(\theta))$.

- (a) (4 points) Write the individual rationality constraint for a type θ consumer.
- (b) (4 points) Write the incentive compatibility constraint of a type θ consumer.
- (c) (4 points) Write (but do not solve) the constrained maximization problem (the second best problem) that the monopolist solves when screening consumers.
- (d) (4 points) Write (but do not solve) the simplified problem of the monopolist in terms of the *virtual surplus*. Note: I am only asking you to write the simplified problem; I am not asking for you to show how you reached it. Thus, you can skip all the steps that were required to obtain the simplification.
- (e) (8 points) For each θ , state the first order condition of the simplified problem of the monopolist.
- (f) (8 points) Solve the first order condition above to obtain the optimal quantity $x(\theta)$.
- (g) (8 points) Obtain the optimal transfer function $T(\theta)$.

Solution

- (a) For the set of contracts $(x(\theta), T(\theta))$, the individual rationality constraint of type θ is:

$$\frac{\theta(1 - (1 - x(\theta))^2)}{2} - T(\theta) \geq 0.$$

- (b) The incentive compatibility constraint for type θ is

$$\frac{\theta(1 - (1 - x(\theta))^2)}{2} - T(\theta) \geq \frac{\theta(1 - (1 - x(\theta'))^2)}{2} - T(\theta'), \quad \forall \theta'$$

Alternatively,

$$\theta \in \arg \max_{\theta'} \left\{ \frac{\theta(1 - (1 - x(\theta'))^2)}{2} - T(\theta') \right\}$$

(c) The monopolist solves:

$$\begin{aligned} \max_{(x(\theta), T(\theta))} \quad & \left\{ \int_{\theta_L}^{\theta_H} (T(\theta) - cx(\theta)) \left(\frac{d\theta}{\theta_H - \theta_L} \right) \right\} \\ \text{subject to:} \quad & \frac{\theta(1 - (1 - x(\theta))^2)}{2} - T(\theta) \geq 0 \quad \forall \theta, \\ & \frac{\theta(1 - (1 - x(\theta))^2)}{2} - T(\theta) \geq \frac{\theta(1 - (1 - x(\theta'))^2)}{2} - T(\theta'), \quad \forall \theta' \text{ and } \forall \theta, \\ & x(\theta) \geq 0, \forall \theta. \end{aligned}$$

(d) Note that the inverse hazard rate for the uniform distribution $(1 - F(\theta))/f(\theta) = \theta_H - \theta$. Then, the simplified monopolist's problem is:

$$\begin{aligned} \max_{x(\theta)} \quad & \left\{ \int_{\theta_L}^{\theta_H} \left(\frac{\theta(1 - (1 - x(\theta))^2)}{2} - cx(\theta) - (\theta_H - \theta) \left(\frac{(1 - (1 - x(\theta))^2)}{2} \right) \right) \left(\frac{d\theta}{\theta_H - \theta_L} \right) \right\} \\ \text{subject to:} \quad & x(\theta) \text{ increasing and } x(\theta) \geq 0, \forall \theta. \end{aligned}$$

(e) For each θ the first order condition is:

$$\theta(1 - x(\theta)) - c - (\theta_H - \theta)(1 - x(\theta)) = 0.$$

(f) Solving for the first order condition yields:

$$(1 - x(\theta))(\theta - (\theta_H - \theta)) = c \Rightarrow 1 - x(\theta) = \frac{c}{2\theta - \theta_H}$$

Thus, when taking into consideration the possibility of corner solutions, the optimal $x(\theta)$ is

$$x(\theta) = \max \left\{ 1 - \frac{c}{2\theta - \theta_H}, 0 \right\}$$

We can be a bit more explicit about the solution. Let $\theta^* = \frac{c + \theta_H}{2}$ be the solution to the equation:

$$1 - \frac{c}{2\theta^* - \theta_H} = 0$$

Then the optimal quantity is

$$x(\theta) = \begin{cases} 0 & \text{if } \theta < \theta^* \\ 1 - \frac{c}{2\theta - \theta_H} & \text{if } \theta \geq \theta^* \end{cases}$$

(g) Transfers are given by

$$T(\theta) = \frac{\theta(1 - (1 - x(\theta))^2)}{2} - \int_{\theta_L}^{\theta} x(\tau) d\tau$$

Simplifying a little yields

$$T(\theta) = \begin{cases} 0 & \text{if } \theta < \theta^* \\ \frac{\theta}{2} \left(1 - \left(\frac{c}{2\theta - \theta_H} \right)^2 \right) - \int_{\theta^*}^{\theta} \left(1 - \frac{c}{2\tau - \theta_H} \right) d\tau & \text{if } \theta \geq \theta^* \end{cases}$$

2. A risk neutral principal hires a risk averse agent to undertake a project, which has a fixed cost of $I \geq 0$ for the principal. The project is either successful, in which case the value of the project for the principal is 20, or it is a failure, in which case the value is 4. The agent chooses to put in either high or low effort, i.e. $e \in \{0, 1\}$ where $e = 0$ corresponds to low effort, and $e = 1$ corresponds to high effort. If he puts in high effort, then the probability of success is 0.75, and if he chooses low effort, the probability of success is 0.25. The principal cannot observe the agent's effort but observes whether the outcome is a success or a failure. Therefore, feasible contracts are payments to the agent in case of success or failure, i.e., (w_s, w_f) . Therefore, the principal's payoff is $20 - w_s$, in case of success, and $4 - w_f$, in case of failure. Assume that the principal's payoff if she does not invest in the project is

zero.

The agent has a utility function given by $\ln(w) - c(e)$, where $c(1) = \ln(4)$ and $c(0) = 0$. Also assume that the reservation utility of the agent (i.e., the payoff he gets if he does not accept the contract offer) is zero.

Facts (Some properties about $\ln$ and e). Here are some useful facts:

- $\lim_{x \rightarrow 0} \ln(x) = -\infty$
- $\ln(e) = 1$, $\ln(4) > 0$, $a \ln(x) = \ln(x^a)$, $\ln(a) + \ln(b) = \ln(ab)$, $e^{\ln(x)} = x$.
- $\frac{d \ln(x)}{dx} = \frac{1}{x}$

- (a) (12 points) What is the optimal contract that induces high effort?
- (b) (12 points) What is the optimal contract that induces low effort?
- (c) (4 points) What effort level is optimal to induce?
- (d) (12 points) Assume that it is optimal to induce high effort. For what range of I , the principal will find it optimal to hire the agent? How about if the principal could observe and contract on effort?

Solution

- (a) The principal's problem is

$$\begin{aligned} \max_{w_s, w_f} \quad & 0.75(20 - w_s) + 0.25(4 - w_f) - I \\ \text{subject to} \quad & 0.75 \ln(w_s) + 0.25 \ln(w_f) - \ln(4) \geq 0 \\ & 0.75 \ln(w_s) + 0.25 \ln(w_f) - \ln(4) \geq 0.25 \ln(w_s) + 0.75 \ln(w_f) \end{aligned}$$

Note that the participation constraint implies that $w_f, w_s > 0$, whereas the incentive compatibility constraint implies that $w_s > w_f$. Thus, we don't need to add non-negativity constraints. The Lagrangian is given by

$$\mathcal{L} = 0.75(20 - w_s) + 0.25(4 - w_f) - I + \mu[0.75 \ln(w_s) + 0.25 \ln(w_f) - \ln(4)] + \lambda[0.5(\ln(w_s) - \ln(w_f)) - \ln(4)]$$

The first order conditions with respect to w_s and w_f are

$$-0.75 + \frac{0.75\mu}{w_s} + \frac{0.5\lambda}{w_s} = 0 \tag{1}$$

$$-0.25 + \frac{0.25\mu}{w_f} - \frac{0.5\lambda}{w_f} = 0 \tag{2}$$

Condition (2) together with $w_f > 0$ and $\lambda \geq 0$ imply that $\mu > 0$. Therefore the participation constraint is binding. Adding up (1) and (2) yields $\mu = 0.75w_s + 0.25w_f$. substituting this (2) yields that $\lambda = \frac{3}{8}(w_s - w_f) > 0$ which implies that the IC constraint is also binding. Therefore, we can use the PC and IC constraints as equalities to solve for w_f and w_s as

$$w_f = \frac{1}{2}, \quad w_s = 8$$

- (b) The principal's problem is

$$\begin{aligned} \max_{w_s, w_f} \quad & 0.25(20 - w_s) + 0.75(4 - w_f) - I \\ \text{subject to} \quad & 0.25 \ln(w_s) + 0.75 \ln(w_f) \geq 0 \\ & 0.25 \ln(w_s) + 0.75 \ln(w_f) \geq 0.75 \ln(w_s) + 0.25 \ln(w_f) - \ln(4) \end{aligned}$$

Note that the participation constraint implies that $w_f, w_s > 0$. Thus, we do not need to add non-negativity constraints.

The Lagrangian is

$$\mathcal{L} = 0.25(20 - w_s) + 0.75(4 - w_f) - I + \mu[0.25 \ln(w_s) + 0.75 \ln(w_f)] + \lambda[0.5(\ln(w_f) - \ln(w_s)) + \ln(4)]$$

The first order conditions with respect to w_s and w_f are

$$-0.25 + \frac{0.25\mu}{w_s} - \frac{0.5\lambda}{w_s} = 0 \quad (3)$$

$$-0.75 + \frac{0.75\mu}{w_f} + \frac{0.5\lambda}{w_f} = 0 \quad (4)$$

Condition (3), $\lambda \geq 0$ and $w_s > 0$ imply that $\mu > 0$. Therefore the participation constraint is binding. Again, adding up (3) and (4) yields $\mu = 0.25w_s + 0.75w_f$. Substituting this into (3) yields that $\lambda = \frac{3}{8}(w_f - w_s)$. Since $\lambda \geq 0$, the previous result implies that $w_f \geq w_s$.

We will consider two cases. Suppose that $w_f > w_s$ so that $\lambda > 0$. Then the IC constraint also binds. Using the binding PC and IC constraints as equalities to solve for w_f and w_s yields once again that $w_s = 8$ and $w_f = 1/2$; however, this is incompatible with $\lambda > 0$ so we get a contradiction. Thus, we must have $w_f = w_s$ which implies that $\lambda = 0$.

The fact that $\lambda = 0$ and $w_f = w_s$, together with the binding PC constraint implies that $w_s = w_f = 1$.

(c) Expected profit under high effort is $9.875 - I$, whereas under low effort is $7 - I$. Clearly, it is optimal to induce high effort.

(d) The principal will hire an agent if $9.875 - I \geq 0$ or $I \leq 9.875$.

If the principal could observe effort, then the only constraint that matters is the PC constraint. Therefore, if the principal wants to induce high effort, her problem is

$$\begin{aligned} \max_{w_s, w_f} \quad & 0.75(20 - w_s) + 0.25(4 - w_f) - I \\ \text{subject to} \quad & 0.75 \ln(w_s) + 0.25 \ln(w_f) - \ln(4) \geq 0 \end{aligned}$$

Once again, notice that the PC constraint implies that w_s and w_f are strictly positive. The Lagrangian is given by

$$\mathcal{L} = 0.75(20 - w_s) + 0.25(4 - w_f) - I + \mu[0.75 \ln(w_s) + 0.25 \ln(w_f) - \ln(4)]$$

The first order conditions with respect to w_s and w_f are

$$-0.75 + \frac{0.75\mu}{w_s} = 0 \quad (5)$$

$$-0.25 + \frac{0.25\mu}{w_f} = 0 \quad (6)$$

From this we conclude that $\mu = w_f = w_s > 0$ and the PC constraint binds. Therefore

$$\ln(w^*) = \ln(4) \Rightarrow w^* = w_s = w_f = 4.$$

Expected profit is equal to $12 - I$. As before, expected profit under low effort is $7 - I$ (because in the second best only the PC constraint binds, so the solution to the first-best low effort is the same). Therefore, when effort is observable, inducing high effort is optimal (no need to show this since I asked you just to assume it) and the principal finds it optimal to hire an agent if $I \leq 12$. This implies that projects that cost $I \in (9.875, 12)$ are not undertaken due to moral hazard.

3. A risk neutral landowner is going to offer a tenancy contract to a financially constrained tenant. The crop yield is either high, q_h , or low q_l , with $q_h > q_l \geq 0$. The tenant chooses to put in either high or low effort. If he puts in high effort, then the probability of high output is p_h , and if he chooses low effort, the probability of high output is p_l , with $p_h > p_l > 0$. The landlord cannot observe the tenant's effort but observes whether the output is low or high, and hence feasible contracts are transfers to the tenant in case of high and low outputs, i.e., (t_h, t_l) . Therefore, the landlord's payoff is $q_h - t_h$ if output is high; and $q_l - t_l$ if output is low. If the tenant receives a transfer t and puts in high effort, then his

payoff is $t - c$, where $c > 0$ is the cost of high effort. If he chooses low effort, then his payoff is simply t . The tenant has limited liability so that $t_l \geq 0$ and $t_h \geq 0$. Also assume that the reservation utility (i.e., the payoff he gets if he does not accept the contract offer) of the tenant is zero. *Hint: This problem has linear profits and utilities, therefore, differentiation will not help you much. It is much better to try to think through this problem without taking derivatives at all. Try to check which constraints need to be binding. A very good way to solve this problem would be to use a graphical approach.*

- (a) (9 points) What is the optimal contract that induces high effort?
- (b) (9 points) What is the optimal contract that induces low effort?
- (c) (4 points) What effort level is optimal to induce?
- (d) (9 points) Assume that it is optimal to induce high effort. Suppose that at cost $p > 0$, the landlord can monitor and learn with certainty whether the agent puts in high or low effort. For what range of p , the landlord will find it optimal to monitor?
- (e) (9 points) Now suppose that there is a functioning credit market so that there are no limited liability constraints anymore. Find the landlord's expected profit under the optimal contract and compare it with her expected profit from part (a).

Solution

(a) The landlord's problem is

$$\begin{aligned}
 \max_{t_h, t_l} \quad & p_h(q_h - t_h) + (1 - p_h)(q_l - t_l) \\
 \text{subject to} \quad & p_h t_h + (1 - p_h)t_l - c \geq 0 & \text{(PC)} \\
 & p_h t_h + (1 - p_h)t_l - c \geq p_l t_h + (1 - p_l)t_l & \text{(IC)} \\
 & t_l \geq 0 & \text{(LL}_l\text{)} \\
 & t_h \geq 0 & \text{(LL}_h\text{)}
 \end{aligned}$$

Let me rewrite the IC constraint as

$$(p_h - p_l)(t_h - t_l) - c \geq 0$$

Firstly, the IC constraint implies that $t_h > t_l$. Therefore, LL_l implies LL_h , and hence, the latter limited liability constraint can be ignored. Secondly, either the PC constraint or the LL_l constraint must bind. Otherwise, the landlord can decrease t_l and t_h by the same amount without violating any of the constraints and increase profits.

We will now consider two cases. Suppose first that PC binds at the solution, i.e.,

$$t_h = \frac{c - (1 - p_h)t_l}{p_h}$$

Therefore,

$$\begin{aligned}
 (p_h - p_l)(t_h - t_l) - c &= (p_h - p_l) \left(\frac{c - (1 - p_h)t_l}{p_h} - t_l \right) - c \\
 &= \frac{p_h - p_l}{p_h} (c - t_l) - c \\
 &= -\frac{p_h - p_l}{p_h} t_l - \frac{p_l}{p_h} c < 0
 \end{aligned}$$

Thus, the IC constraint is violated so we have reached a contradiction. Therefore, the LL_l must bind, i.e., $t_l = 0$. This in turn, implies that the IC constraint must bind. Otherwise, since the PC constraint does not bind either, the landlord could increase profits by decreasing t_h by some small amount without violating any of the constraints.

Therefore, the optimal contract is

$$t_l = 0, \quad t_h = \frac{c}{p_h - p_l}$$

(b) The landlord's problem is

$$\begin{aligned}
& \max_{t_h, t_l} && p_l(q_h - t_h) + (1 - p_l)(q_l - t_l) \\
\text{subject to} &&& p_l t_h + (1 - p_l)t_l \geq 0 && \text{(PC)} \\
&&& p_l t_h + (1 - p_l)t_l \geq p_h t_h + (1 - p_h)t_l - c && \text{(IC)} \\
&&& t_l \geq 0 && \text{(LL}_l\text{)} \\
&&& t_h \geq 0 && \text{(LL}_h\text{)}
\end{aligned}$$

Let me rewrite the IC constraint as

$$(p_h - p_l)(t_l - t_h) + c \geq 0$$

First suppose that $t_h > 0$. A strictly positive t_h implies that the PC constraint holds with strict inequality, i.e., is not binding. But if $t_h > 0$ this implies that the principal can lower it a little bit without violating any constraints and increase its profits. Therefore, we must have that $t_h = 0$. With the previous result, we can similarly conclude that if $t_l = 0$, then the principal can reduce it by some small amount without violating any of the constraints and increase her profits. Thus, $t_l = 0$.

We conclude that the optimal contract in this case is $t_l = t_h = 0$. Note that the limited liability constraints do not impose any burden with respect to the problem of the landlord without them, the solution would still be the same.

(c) The expected profit under high effort is

$$p_h \left(q_h - \frac{c}{p_h - p_l} \right) + (1 - p_h)q_l$$

and under low effort is

$$p_l q_h + (1 - p_l)q_l$$

The expected profit under high effort is higher if and only if

$$(p_h - p_l)(q_h - q_l) \geq \frac{c}{p_h - p_l}$$

If the previous condition holds, it is optimal to induce high effort, otherwise it is optimal to induce low effort.

(d) If the landlord is able to monitor by paying for the monitoring technology, then the only constraint that matters is the PC constraint, since the landlord can punish the tenant if he puts in an effort level that she does not desire to induce. Therefore, if she wants to induce high effort, the landlord's problem is

$$\begin{aligned}
& \max_{t_h, t_l} && p_h(q_h - t_h) + (1 - p_h)(q_l - t_l) - p \\
\text{subject to} &&& p_h t_h + (1 - p_h)t_l - c \geq 0 && \text{(PC)} \\
&&& t_l \geq 0 && \text{(LL}_l\text{)} \\
&&& t_h \geq 0 && \text{(LL}_h\text{)}
\end{aligned}$$

At the solution, the PC constraint must bind. If it doesn't, then $t_h > 0$ or $t_l > 0$, and hence the landlord can decrease t_l or t_h without violating any of the constraints and increase profit. Therefore,

$$p_h t_h + (1 - p_h)t_l - c = 0$$

together with $t_l \geq 0$ and $t_h \geq 0$ characterize the *solution set* to the landlord's problem. The expression in the display is the expected cost of any solution (t_l, t_h) that satisfies the previous conditions. Therefore, we can substitute this expected cost into the objective function to get the expected profit of the landlord:

$$p_h q_h + (1 - p_h)q_l - c - p$$

Therefore, monitoring is optimal if and only if

$$p_h q_h + (1 - p_h) q_l - c - p \geq p_h \left(q_h - \frac{c}{p_h - p_l} \right) + (1 - p_h) q_l$$

The previous expression is equivalent to

$$p \leq \frac{c p_l}{p_h - p_l}$$

(e) The landlord's problem is now

$$\begin{aligned} \max_{t_h, t_l} \quad & p_h(q_h - t_h) + (1 - p_h)(q_l - t_l) \\ \text{subject to} \quad & p_h t_h + (1 - p_h) t_l - c \geq 0 \quad (\text{PC}) \\ & p_h t_h + (1 - p_h) t_l - c \geq p_l t_h + (1 - p_l) t_l \quad (\text{IC}) \end{aligned}$$

At the solution, the PC constraint must bind, otherwise, the landlord could decrease t_l and t_h by the same small amount and increase her profit. Therefore,

$$p_h t_h + (1 - p_h) t_l - c = 0$$

Substituting this into the expected profit function we get

$$p_h q_h + (1 - p_h) q_l - c$$

The expected profit of the landlord when the tenant was financially constrained was

$$(p_h - p_l)(q_h - q_l) \geq \frac{c}{p_h - p_l}$$

Since $\frac{p_h}{p_h - p_l} > 1$ the expected profit when the tenant is not financially constrained is higher.